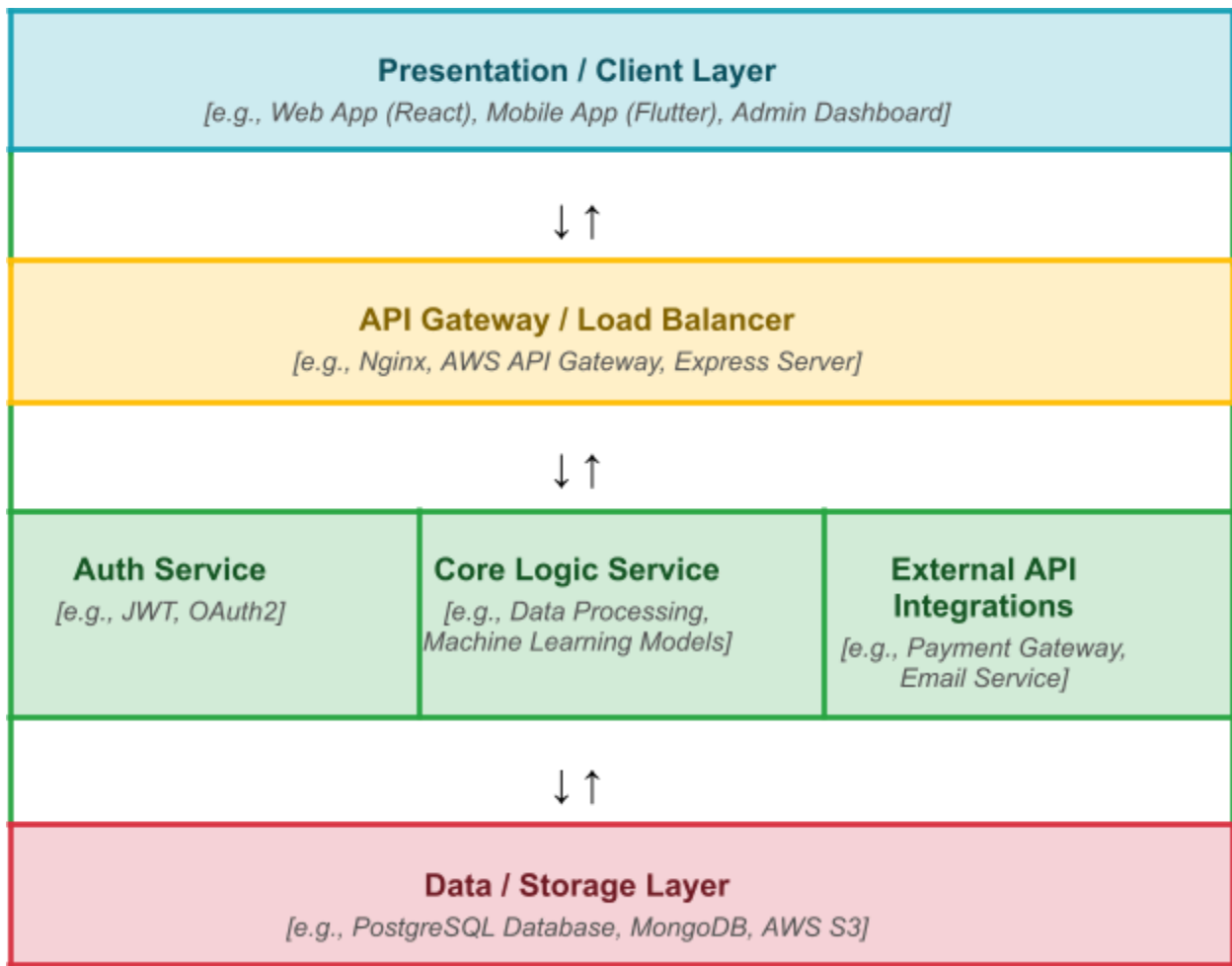


Solution Architecture

Date	26 june 2026
Team ID	SWIID-2026-1243
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

The **Solution Architecture Diagram** illustrates the high-level architecture of **EduGenie**, an AI-powered educational assistant. The system follows a layered architecture consisting of the **Presentation Layer**, **Application Layer**, and **Data Layer**. The Presentation Layer provides a responsive web interface built with **HTML**, **CSS**, and **JavaScript** for user interaction. The Application Layer is powered by **FastAPI**, which processes user requests and integrates **Google Gemini 1.5 Pro** and **LaMini-Flan-T5** to perform intelligent question answering, concept explanation, quiz generation, text summarization, and personalized learning recommendations. The Data Layer manages user information, learning history, application logs, and educational content using a lightweight database. This architecture ensures a scalable, efficient, and user-friendly AI-powered learning platform.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive web interface where users can ask questions, generate quizzes, summarize text, and receive learning recommendations.	HTML5, CSS3, JavaScript
[API Gateway]	Handles client requests, routes them to the appropriate API endpoints, validates input, and returns AI-generated responses.	FastAPI, Uvicorn

<i>[Microservices]</i>	Processes educational requests, performs prompt engineering, integrates AI models, and generates responses for question answering, concept explanation, quiz generation, summarization, and learning recommendations.	Python, Google Gemini 1.5 Pro, LaMini-Flan-T5, Transformers, Torch
<i>[Database]</i>	Stores user information (optional), quiz history, learning progress, application logs, and configuration data.	SQLite / PostgreSQL